

Assembling Data Sets

FATEMEH RIAHI
BCIT



Data Munging

Good data scientists spend most of their time cleaning and formatting data.

The rest spend most of their time complaining there is no data available.

Data munging or *data wrangling* is the art of acquiring data and preparing it for analysis.

Languages for Data Science

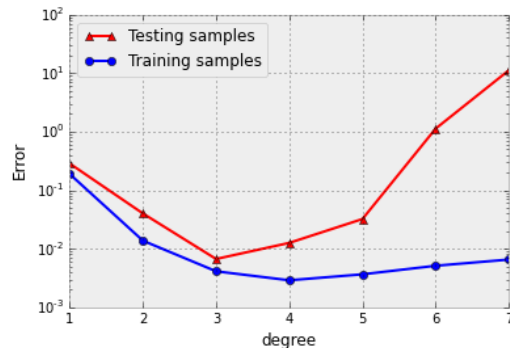
- *Python*: contains libraries and features (e.g regular expressions) for easier munging.
- *R*: programming language of statisticians.
- *Matlab*: fast and efficient matrix operations.
- *Java/C*: language for Big Data systems.
- *Mathematica/Wolfram Alpha*: symbolic math.
- *Excel*: bread and butter tool for exploration.

Notebook Environments

Mixing code, data, computational results, and text are essential for projects to be:

- reproducible
- tweakable
- documented.

```
In [40]: degrees = range(1, 8)
errors = np.array([regressor3(d) for d in degrees])
plt.plot(degrees, errors[:, 0], marker='^', c='r', label='Testing samples')
plt.plot(degrees, errors[:, 1], marker='o', c='b', label='Training samples')
plt.yscale('log')
plt.xlabel("degree"); plt.ylabel("Error")
plt.legend(loc='best')
```



By sweeping the degree we discover two regions of model performance:

- **Underfitting** (degree < 3): Characterized by the fact that the testing error will get lower if we increase the model capacity.
- **Overfitting** (degree > 3): Characterized by the fact the testing will get higher if we increase the model capacity. Note, that the training error is getting lower or just staying the same!

Data Pipelines

Notebooks make it easier to maintain data pipelines, the sequence of processing steps from start to finish.

Expect to have to redo your analysis from scratch, so build your code to make it possible.



Standard Data Formats

Historically, computer scientists would rather share a toothbrush than a data format.

But accepted standards are available:

- *CSV files*: for tables like spreadsheets
- *XML*: for structured but non-tabular data.
- *JSON*: Javascript Object Notation for APIs.
- *SQL databases*: for multiple related tables.
- *Pickle*: serialize data structures in Python.

Where Does Data Come From?

The critical issue in any modelling project is finding the right data set.

This will certainly be the case for your projects!

Large data sets often come with valuable **metadata**: e.g. book titles, image captions, Wikipedia edit history...

Repurposing metadata requires imagination.

Sources of Data

- Proprietary data sources
- Government data sets
- Academic data sets
- Web search
- Sensor data
- Crowdsourcing
- Sweat equity

<https://toolbox.google.com/datasetsearch>

Proprietary Data Sources

Facebook, Google, Amazon, Blue Cross, etc. have exciting user/transaction/log data sets.

Most organizations have/should have internal data sets of interest to their business.

Getting outside access is usually impossible.

Companies sometimes release rate-limited APIs, including Twitter and Google.

Decreasing Real Public Data



The commercial promise of large language models (LLMs) has prompted once open data providers like Twitter and Reddit to tighten up.

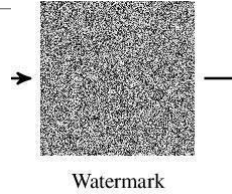
The use of AI-generated text and images on the web complicate the task of training improved models.

Recognizing Generated Text/Images

Watermarking adds subliminal data to an object to recognize it as original.

Adding low-order bit patterns can watermark them: but this can be easily masked away.

ChatGPT *may* watermark text by using a pseudo-random generator for decoding.



Government Data Sources

- City, State, and Federal governments are increasingly committed to open data.
- Data.gov has over 250,000 open data sets!
- The Freedom of Information Act (FOI) enables you to ask if something is not open.
- Preserving privacy is often the big issue in whether a data set can be released.



Academic Data Sets

-
- Making data available is now a requirement for publication in many fields.
 - Expect to be able to find economic, medical, demographic, and meteorological data if you look hard enough.
 - Track down from relevant papers, and ask.
 - Google topic and “Open Science” or “data”

Web Scraping

BeautifulSoup



Scraping is the fine art of stripping text/data from a webpage.

Python Libraries (BeautifulSoup, Selenium), exist to help parse/scrape the web, but search:

- Are APIs available from the source?
- Did someone previously write a scraper?

Terms of service limit what you can legally do.

Sensor Data Logging

The “Internet of Things” can do amazing things:

- Image/video data can do many things: e.g. measuring the weather using Flickr images.
- Measure earthquakes using accelerometers in cell phones.
- Identify traffic flows through GPS on taxis.

Build logging systems: storage is cheap!

Crowdsourcing

Many amazing open data resources have been built up by teams of contributors:

- Wikipedia/WikiData
- IMDB

Crowdsourcing platforms like Amazon Turk enable you to pay for armies of people to help you gather data, like human annotation.

Sweat Equity

But sometimes you must work for your data instead of stealing it.

Much historical data still exists only on paper or PDF, requiring manual entry/curation.

At one record per minute, you can enter 1,000 records in only two work days.

Often projects require sweat equity.

Cleaning Data: Garbage In, Garbage Out

Many issues arise in ensuring the sensible analysis of data from the field, including:

- Distinguishing errors from artifacts.
- Data compatibility / unification.
- Imputation of missing values.
- Estimating unobserved (zero) counts.
- Outlier detection.

Errors vs. Artifacts

-
- Data **errors** represent information that is fundamentally lost in acquisition.
 - **Artifacts** are systematic problems arising from processing done to data.

The key to detecting artifacts is the **sniff test**, examining the product closely enough to get a whiff of something bad.

First-time Scientific Authors by Year?

In a bibliographic study, we analyzed PubMed data to identify the year of first publication for the 100,000 most frequently cited authors.

What *should* the distribution of new top authors by year look like?

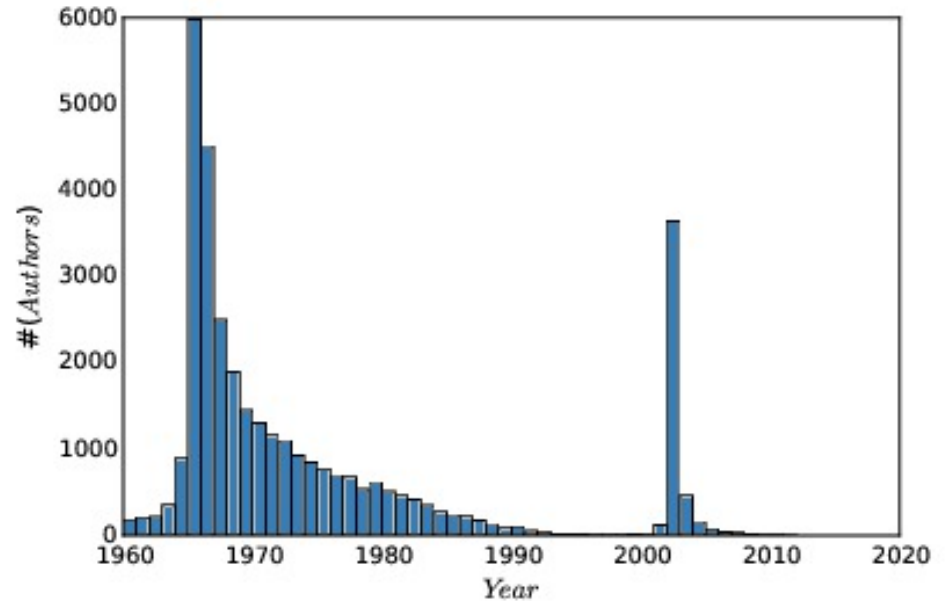
It is important to have a preconception of any result to help detect anomalies.

Might this be Right?

A student once tried to foist this off on me.

What artifacts do you see?

What possible explanations could cause them?

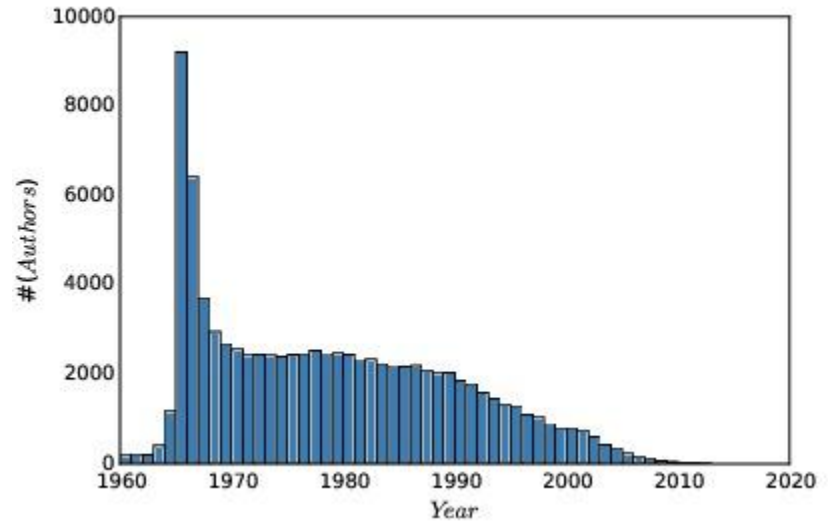


Mystery Solved!

Pubmed used author first names starting in 2002.

F Riahi became
Fatemeh Riahi

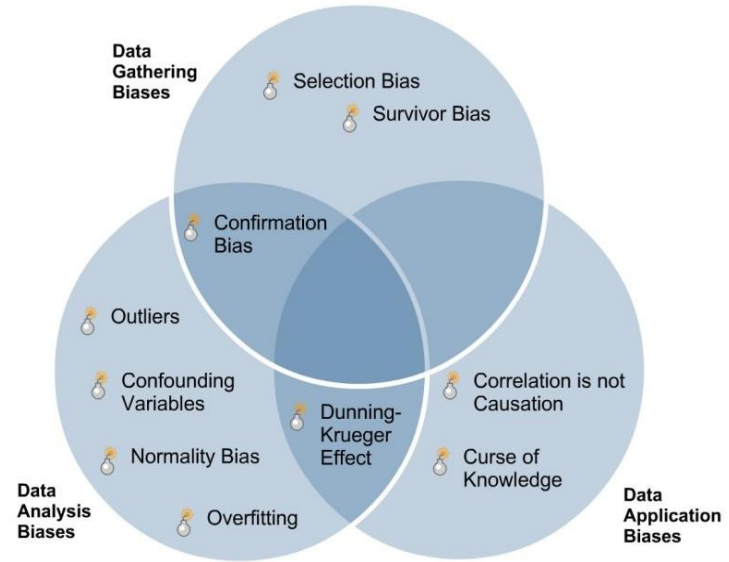
Data cleaning gets rid of such artifacts.



Data Cleaning

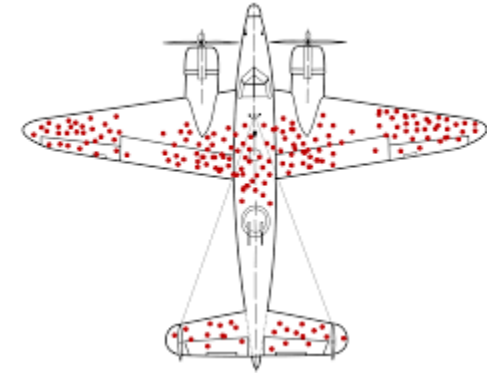
Biases in Sampling and Analysis

- Data bias occurs when data is inaccurate, incomplete, or missing, which can lead to an inaccurate representation of a population.
- The best way to clean data is to eliminate systematic biases in data collection.

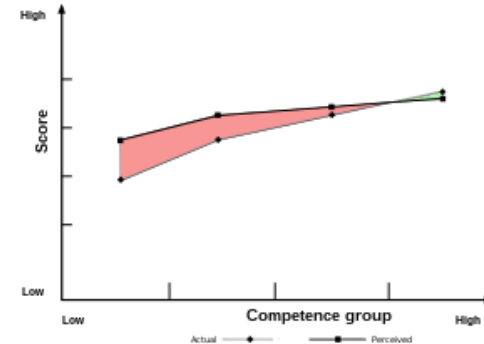


Classic Sampling/Analysis Biases

- *Selection bias* – over/under sample certain groups.
- *Survivor bias* – by selecting at the end, you undersample who exits early.
- *Confirmation bias* – people stop sampling / analyzing when the results confirm expectation.
- *Dunning-Kruger effect* – poor performers over-estimate their level of performance



Dunning-Kruger Effect



Cleaning Data: Garbage In, Garbage Out

Many issues arise in ensuring the sensible analysis of data from the field, including:

- Distinguishing errors from artifacts.
- Data compatibility / unification.
- Imputation of missing values.
- Estimating unobserved (zero) counts.
- Outlier detection.

Errors vs. Artifacts

-
- Data **errors** represent information that is fundamentally lost in acquisition.
 - **Artifacts** are systematic problems arising from processing done to data.

The key to detecting artifacts is the **sniff test**, examining the product closely enough to get a whiff of something bad.

Data Compatibility

Data needs to be carefully massaged to make “apple to apple” comparisons:

- Unit conversions
- Number / character code representations
- Name unification
- Time/date unification
- Financial unification

Unit Conversions



Remember the Mars Climate Orbiter incident from 1999?

NASA's Mars Climate Orbiter exploded in 1999 due to a metric-to-English conversion issue.

- Even sticking to the metric system has potential inconsistencies: cm, m, km?
- Bimodal distributions can indicate trouble
- Z-scores are dimensionless quantities.

Vigilance in data integration is essential.

Normalization and Z-scores

It is critical to normalize different variables to make their range/distribution comparable.

Z-scores are computed:

Z-scores of height measured in inches is the same as height measured in miles.

Your biggest analysis sins will come in using unnormalized variables for analysis!

$$Z_i = (X_i - \bar{X})/\sigma$$

Z-score Examples

Z-scores have mean 0 and sigma=1.

Thus Z-scores of different variables are of comparable magnitude.

The sign identifies if it is above/below the mean.

$$\begin{array}{ll} \mu(B) = 21.9 & \sigma(B) = 1.92 \\ \mu(Z) = 0 & \sigma(Z) = 1 \end{array}$$

B	19	22	24	20	23	19	21	24	24	23
Z	-1.51	0.05	1.09	-0.98	0.57	-1.51	-0.46	1.09	1.09	0.57

Number / Character Representations

The Ariane 5 rocket exploded in 1996 due to a bad 64-bit float to 16-bit integer conversion.

- Measurements should be decimal numbers
- Counts should be integers.
- Fractional quantities should be decimal, not (q,r) like (pounds,oz) or (feet,inches).

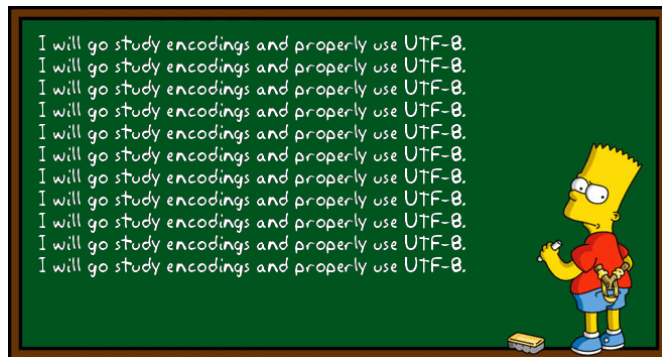


Character Representations

A particularly nasty cleaning issue in textual data is unifying character code representations:

- ISO 8859-1 is a single byte code for ASCII
- UTF-8 is a multibyte encoding for all Unicode characters.

Unicode font, UTF8 format	Unicode font, XXX... format
搜索简体中文网页	????????
Recherche avancée	Recherche avancée
網路畫廊，含中、港、澳參展作品	??????????????
โทรศํพณภว	??????????????
ウェブ全体から	??????
kehren Sie zur Suche zurück	kehren Sie zur Suche zurück
Сделайте Google стартовой	???????? Google ???????
إخدوقت بحث أقل ووقت مطالعة أطول	?????? ??? ??? ??? ?????? ????



Name Unification

I appear on the web as: *(Steve/Steven/Stephen/S.) (S./Sol/_)*
(Skiena/Skeina/Skienna)

- Use simple transformations to unify names, like lower case, removing middle names, etc.
- Consider phonetic hashing methods like Soundex and Metaphone.

Tradeoff between false positives and negatives.

Time / Date Unification

Aligning temporal events from different datasets/systems can be problematic.

- Use Coordinated Universal Time (UTC), a modern standard subsuming GMT.
- Financial time series are tricky because of weekends and holidays: how do you correlate stock prices and temperatures?

September 1752						
Su	M	Tu	W	Th	F	Sa
-	-	1	2	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30

Financial Unification

- Currency conversion uses exchange rates.
- Correct stock prices for splits and dividends.
- Use returns / percentage change instead of absolute price changes.
- The time value of money needs correction for inflation for fair long-term comparisons.

Why do stock/oil prices correlate over 30 years?

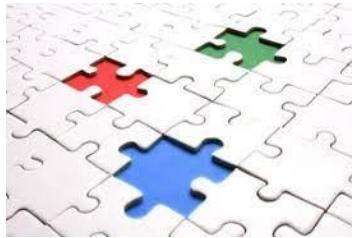


Dealing with Missing Data

An important aspect of data cleaning is properly representing missing data:

- What is the year of death of a living person?
- What about a field left blank or filled with an obviously outlandish value?
- The frequency of events too rare to see?

Setting such values to zero is generally wrong



Imputing Missing Values

With enough training data, one might drop all records with missing values, but we may want to use the model on records with missing fields

Often it is better to estimate or **impute** missing values instead of leaving them blank.

A good guess for your death year is $\text{birth} + 80$.

Imputation Methods

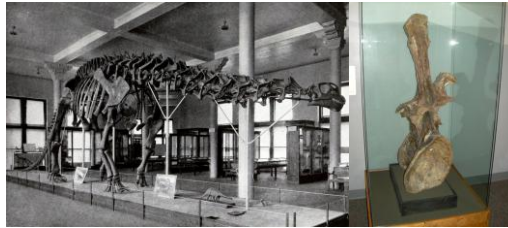
- *Mean value imputation* - leaves mean same.
- *Random value imputation* - repeatedly selecting random values permits statistical evaluation of the impact of imputation.
- *Imputation by interpolation* - using linear regression to predict missing values works well if few fields are missing per record.

Outlier Detection

The largest reported dinosaur vertebra is 50% larger than all others: presumably a data error.

- Look critically at the maximum and minimum values for all variables.
- Normally distributed data should not have large outliers, k *sigma* from the mean.

Fix why you have an outlier. Don't just delete.



Detecting Outliers

-
- Visually, it is easy to detect outliers, but only in low dimensional spaces.
 - It can be thought of as an unsupervised learning problem, like clustering.
 - Points which are far from their cluster center are good candidates for outliers

Delete Outliers Prior to Fitting?

-
- Deleting outliers prior to fitting **can yield better models**, e.g. if these points correspond to measurement error.
 - Deleting outliers prior to fitting **can yield worse models**, e.g. if you are simply deleting points which are not explained by your simple model.

Resources

I would like to thank **Dr. Steven S. Skiena**, **Dr. Christian Kästner**, and **Chip Huyen** for the valuable resources they have provided through their coursework in machine learning and data science. Their materials have been instrumental in shaping my teaching in these areas.

- The Data Science Design Manual, Steven S. Skiena
- Learning From Data, by Yaser S. Abu-Mostafa, Malik Magdon-Ismael, Hsuan-Tien Lin
- Machine Learning in Production: From Models to Products by Christian Kästner
- Designing Machine Learning Systems, Chip Huyen